

Project: TradeNet - Public Trade & Commerce Governance System

1. Introduction

TradeNet is a web-based platform designed for commerce ministries, trade boards, and government agencies to manage public trade, market regulation, and compliance. It enables businesses to register, file trade licenses, and track transactions, while administrators and officers can monitor compliance, manage trade programs, and generate reports.

The system supports workflows for business registration, trade licensing, transaction tracking, compliance monitoring, and analytics. It ensures transparency and accountability by maintaining audit trails, dashboards, and performance metrics across trade governance.

Actors / Users

- **Business/Trader:** Registers, applies for trade licenses, tracks transactions.
- **Trade Officer:** Validates licenses, monitors transactions, updates compliance status.
- **Program Manager:** Oversees trade programs, monitors revenue, tracks performance.
- **Administrator:** Configures workflows, manages users, oversees reporting.
- **Compliance Officer:** Ensures adherence to trade policies, audits records.
- **Government Auditor:** Reviews compliance reports, monitors trade program utilization.

2. Module Overview

- 2.1 Identity & Access Management
- 2.2 Business Registration & Profile Management
- 2.3 Trade License & Document Management
- 2.4 Transaction & Market Regulation
- 2.5 Trade Program & Resource Management
- 2.6 Compliance & Audit Management
- 2.7 Reporting & Analytics
- 2.8 Notifications & Alerts

3. Architecture Overview

- **Frontend:** Angular or React for responsive dashboards.
- **Backend:** REST API-based microservices (Identity, Businesses, Licenses, Transactions, Programs, Compliance, Reports).
- **Database:** Relational DB (MySQL/PostgreSQL/SQL Server).

- **Deployment:** Cloud/on-prem with API gateway, WAF, centralized logging.

4. Module Wise Design

4.1 Identity & Access Management

Features

- Secure authentication and role-based access control.
- Audit logging of all actions.

Entities

- **User**(UserID, Name, Role [Business/Officer/Manager/Admin/Compliance/Auditor], Email, Phone, Status)
- **AuditLog**(AuditID, UserID, Action, Resource, Timestamp)

4.2 Business Registration & Profile Management

Features

- Registers businesses and maintains profiles.
- Validates identity and trade documents.

Entities

- **Business**(BusinessID, Name, Type [Trader/Exporter/Importer], Address, ContactInfo, Status)
- **BusinessDocument**(DocumentID, BusinessID, DocType [License/IDProof], FileURI, UploadedDate, VerificationStatus)

4.3 Trade License & Document Management

Features

- Businesses apply for trade licenses.
- Officers validate licenses.
- Track license status and associated documents.

Entities

- **TradeLicense**(LicenseID, BusinessID, Type [Import/Export/Local], IssuedDate, ExpiryDate, Status)
- **LicenseDocument**(DocumentID, LicenseID, DocType [Application/Approval], FileURI, UploadedDate, VerificationStatus)

4.4 Transaction & Market Regulation

Features

- Track trade transactions.
- Monitor market regulation compliance.
- Record transaction details.

Entities

- **Transaction**(TransactionID, BusinessID, Type [Sale/Purchase], Amount, Date, Status)
- **MarketRecord**(RecordID, TransactionID, OfficerID, Notes, Date, Status)

4.5 Trade Program & Resource Management

Features

- Manage trade programs (export promotion, subsidies).
- Track resources and utilization.
- Monitor program outcomes.

Entities

- **TradeProgram**(ProgramID, Title, Description, StartDate, EndDate, Budget, Status)
- **Resource**(ResourceID, ProgramID, Type [Funds/Materials], Quantity, Status)

4.6 Compliance & Audit Management

Features

- Ensure adherence to trade policies.
- Maintain compliance records.
- Conduct audits with findings.

Entities

- **ComplianceRecord**(ComplianceID, EntityID, Type [License/Transaction/Program], Result, Date, Notes)
- **Audit**(AuditID, OfficerID, Scope, Findings, Date, Status)

4.7 Reporting & Analytics

Features

- Generate dashboards for licenses, transactions, programs, and compliance.
- Provide analytics on trade program efficiency.

Entities

- **Report**(ReportID, Scope [License/Transaction/Program], Metrics, GeneratedDate)

4.8 Notifications & Alerts

Features

- Send alerts for license approvals, transaction updates, compliance deadlines.
- Notify businesses about trade program updates.

Entities

- **Notification**(NotificationID, UserID, EntityID, Message, Category [License/Transaction/Program/Compliance], Status, CreatedDate)

5. Deployment Strategy

- **Local**: Angular/React frontend, Spring Boot/.NET Core backend, local DB.
- **Production**: Cloud/on-prem deployment, API gateway, WAF, centralized logging.

6. Database Design

Tables & Attributes

- **User**(UserID, Name, Role, Email, Phone, Status)
- **AuditLog**(AuditID, UserID, Action, Resource, Timestamp)
- **Business**(BusinessID, Name, Type, Address, ContactInfo, Status)
- **BusinessDocument**(DocumentID, BusinessID, DocType, FileURI, UploadedDate, VerificationStatus)
- **TradeLicense**(LicenseID, BusinessID, Type, IssuedDate, ExpiryDate, Status)
- **LicenseDocument**(DocumentID, LicenseID, DocType, FileURI, UploadedDate, VerificationStatus)
- **Transaction**(TransactionID, BusinessID, Type, Amount, Date, Status)
- **MarketRecord**(RecordID, TransactionID, OfficerID, Notes, Date, Status)
- **TradeProgram**(ProgramID, Title, Description, StartDate, EndDate, Budget, Status)
- **Resource**(ResourceID, ProgramID, Type, Quantity, Status)
- **ComplianceRecord**(ComplianceID, EntityID, Type, Result, Date, Notes)
- **Audit**(AuditID, OfficerID, Scope, Findings, Date, Status)
- **Report**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, EntityID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Business Portal:** License application, transaction tracking, notifications.
- **Trade Officer Dashboard:** License validation, transaction oversight, compliance tracking.
- **Program Manager Console:** Program oversight, resource monitoring, reporting.
- **Administrator Panel:** Workflow setup, compliance monitoring, reporting.
- **Compliance Officer Console:** Policy monitoring, audits, compliance reporting.
- **Government Auditor Dashboard:** Trade program review, compliance monitoring.

8. Non-Functional Requirements

- **Performance:** Handle 200,000 concurrent users across trade networks.
- **Security:** Role-based access, encrypted data storage, immutable audit logs.
- **Scalability:** Support nationwide rollout across multiple trade programs.
- **Availability:** 99.9% uptime.
- **Maintainability:** Modular microservices, API versioning, automated migrations.
- **Observability:** Centralized logging, KPIs (license approval rate, transaction success rate, compliance adherence).

9. Assumptions & Constraints

- Initial rollout for a single trade program before nationwide expansion.
- Notifications limited to in-app and SMS/email alerts.
- Fully implementable using Java Spring Boot/ASP.NET Core, Angular/React, and relational DB.